CDX NA IG and NASDAQ Under Stress

Managing Drawdown and Recovery Risk with Credit-Market Signals

A Three-State Markov-Switching VAR with Model-Defined Velocity Regimes

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Abstract

This paper examines whether the CDX North American Investment Grade (CDX NA IG) credit default swap index can inform the dynamic sizing of equity exposure for a long-horizon NASDAQ investor over the course of a stress episode, trimming exposure as a correction develops and rebuilding it as the episode resolves, rather than forecasting corrections from calm markets. Using 3,668 daily observations from September 2011 through June 2026 and a 3-state Markov-Switching VAR (MS-VAR), we let the model define stress regimes endogenously from the volatility of CDX spread velocity rather than from imposed level thresholds. The model identifies three regimes by the volatility of the 3-day percentage change (the velocity) of CDX: Low-Stress ($\sigma \approx 2.2\%$), Moderate-Stress ($\sigma \approx 3.0\%$), and High-Stress ($\sigma \approx 7.6\%$).

We establish three results. First, CDX is a powerful and reliable coincident stress indicator: when credit spreads widen sharply, NASDAQ is falling over the same window in roughly 85–91% of cases, and the two markets share a strong, stable negative correlation across all regimes (-0.67 to -0.82).

Second, CDX becomes a statistically significant predictor of NASDAQ specifically in elevated-stress regimes. Across six specifications, the CDX leadership holds universally in Moderate-Stress and at short horizons in High-Stress regimes, but is absent in calm markets. The credit-to-equity channel therefore activates specifically when stress is elevated.

Third, we identify a robust recovery signal that is strengthened by requiring confirmation. A sharp CDX tightening in the High-Stress regime precedes a positive NASDAQ recovery, and a confirmed tightening, defined as a second such move within roughly one month, is markedly more reliable than a single one. Under episode-clustered inference the confirmed signal is highly significant, and it survives out-of-sample validation: with the COVID crash and the 2022 bear market held in-sample, the signal tested only on the ordinary corrections of 2023 to 2026 continues to precede meaningful recoveries. An event study of eleven major corrections from 2011 to 2026 confirms the framework, with the widening signal registering during every event and the confirmed tightening clustering near the eventual trough. We further document that the magnitude of the CDX move is a strong contemporaneous gauge of drawdown depth (Pearson $r = -0.85$), allowing an investor to scale any allocation adjustment to the severity of the credit dislocation.

Together these findings position CDX NA IG as a high-value overlay for a NASDAQ allocation: a reliable coincident barometer of stress, a regime-conditional leading indicator during stressed periods, a quantitative measure of likely drawdown depth, and a confirmed, out-of-sample-validated signal of the subsequent recovery.

Keywords: CDX NA IG, NASDAQ, credit default swap index, dynamic asset allocation, risk overlay, Markov switching VAR, velocity and volatility regimes, Granger causality, drawdown magnitude, recovery signal

JEL: G11, G12, G13, G17, C32, C58

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1. Introduction

Credit derivatives are financial contracts whose value derives from the creditworthiness of an underlying borrower or pool of borrowers. The most important is the credit default swap (CDS), effectively an insurance contract on default: the protection buyer pays a periodic premium, quoted as a spread in basis points, and receives compensation if the reference entity defaults. The spread therefore moves with the market's assessment of default risk, widening as perceived risk rises and tightening as it falls. Because CDS trade continuously in liquid, forward-looking markets and aggregate the views of sophisticated institutional participants, credit spreads have long been regarded as among the most sensitive real-time barometers of financial stress. Related instruments include single-name CDS on individual issuers, standardized CDS indices that bundle many names, and tranched products that slice a portfolio's credit risk into seniority layers.

Among these, standardized indices are the natural choice for gauging market-wide rather than firm-specific stress, since a single number captures the credit risk of a broad cross-section of issuers. The leading families are the CDX indices, referencing North American issuers, and the iTraxx indices, referencing European and Asian issuers. Within the CDX family, the most widely followed benchmark is CDX NA IG, an equally weighted basket of 125 investment-grade North American corporate issuers. We choose it for four reasons. First, it is among the most liquid credit instruments in the world, so its spread reflects genuine, continuously updated information rather than stale marks. Second, being investment-grade, its movements are driven by broad macro-financial conditions rather than by idiosyncratic distress in a handful of high-yield names, making it a cleaner proxy for systemic stress than CDX High Yield. Third, its North American, large-cap composition aligns naturally with the issuer base underlying U.S. equity indices, creating a direct economic link to the market we study. Fourth, it has a long, continuous, and consistently constructed history, enabling robust multi-cycle analysis. The alternatives are less suitable: single-name CDS are dominated by firm-specific noise, and iTraxx references a different economy, which we leave to future work.

The economic intuition motivating this study is that credit markets may act as early-warning systems for broader financial distress. Credit spreads aggregate dispersed private information about default risk, and sophisticated institutional participants, particularly banks that lend to the reference firms and hedge through CDS, may trade credit ahead of repositioning equity portfolios. If this informational advantage is systematic, credit indices should lead equity indices, especially around episodes of stress. This paper subjects that intuition to a rigorous test using CDX NA IG and the NASDAQ Composite, a technology-heavy U.S. equity benchmark whose sensitivity to funding conditions and risk appetite makes it a demanding and relevant test case.

An institutional mechanism may reinforce CDX's informational role. Because corporate bonds are far less liquid than the CDX market, asset managers often adjust credit exposure through liquid CDS indices rather than by trading the underlying bonds directly. In periods of stress, buying or reducing CDX protection allows institutional investors to hedge or restore credit risk quickly and

at lower transaction cost. Since this activity occurs primarily in an over-the-counter institutional market, CDX price movements can reflect shifts in professional risk appetite before these adjustments are fully visible in cash credit or equity markets. This is the informational channel tested in the paper: CDX may aggregate and price institutional stress ahead of broader market adjustment.

This motivates our focus on the investment-grade credit index rather than on equity- or rate-volatility indicators alone. Equity-implied volatility (the VIX) and Treasury-implied volatility (the MOVE index) are valuable stress measures, but each can react to shocks specific to its own market, producing elevated readings at times when broad credit conditions, as summarized by CDX, remain calm. An equity-options repricing or a rates-driven volatility spike need not reflect a genuine, system-wide deterioration in credit. Because the investment-grade CDS index responds primarily to broad macro-financial credit conditions, it offers a comparatively clean read on systemic stress.

Importantly, our perspective throughout is that of a long-horizon investor who wishes to remain invested in the NASDAQ but to use credit-market information to adjust the size of that exposure through periods of stress. We are not proposing a market-timing strategy that exits and re-enters wholesale; the analysis treats CDX as an overlay signal that informs whether to trim, hold, or rebuild equity allocation as a correction develops and then resolves. The practical questions are therefore: when credit signals that stress is real, how should an investor scale equity exposure, and when credit signals that the stress has passed, when should that exposure be rebuilt? Framed this way, the value of CDX lies not in predicting corrections from calm markets but in helping size risk dynamically once stress is underway and in identifying the turn back up.

We make one important methodological departure from the prior literature and from earlier drafts of this work. Rather than imposing fixed spread-level thresholds (for example, 50 and 100 basis points) to define stress regimes, we let the 3-state Markov-Switching VAR define regimes endogenously, based on the volatility of CDX spread velocity; that is, the dispersion of the rate of change of spreads. This change is motivated by a robust empirical observation: stress regimes are characterized by how turbulent the rate of spread change is, not by the absolute level at which spreads sit. A spread of 85 bps reached through a violent multi-day jump is a fundamentally different state from the same 85 bps reached through a slow drift, and only a velocity-based definition captures that distinction.

Our analysis yields three findings, each of practical value. First, CDX is an excellent coincident stress barometer: it moves in lockstep with NASDAQ (same-day correlation near −0.75) and reliably flags that a drawdown is in progress. Second, and more powerfully, CDX becomes a statistically significant leading indicator of NASDAQ in the moderate- and high-stress regimes - a result that holds across every signal specification we test, making it a structural feature of stressed markets. Third, and of greatest practical value, CDX provides a robust recovery signal: a sharp credit tightening during high stress reliably marks the correction bottom and precedes a significant

NASDAQ rebound. CDX thus provides its greatest informational value at two junctures of particular importance to investors: the confirmation of a developing correction and the identification of the subsequent turning point.

1.1 Contribution

We contribute to the credit-equity literature in four ways. First, we introduce regimes defined endogenously by the volatility of spread velocity, in place of level thresholds. Second, we provide a comprehensive robustness analysis across six signal specifications, showing the regime-conditional leadership is not an artifact of any single definition. Third, we cleanly separate coincident confirmation from genuine prediction, correcting a conflation common in the applied literature. Fourth, we document the asymmetry between the peak and the trough of an equity correction and translate it into a concrete allocation overlay for a held NASDAQ position, showing that CDX magnitude scales the defensive trim while a high-stress credit tightening signals when to rebuild exposure in equities.

1.2 Structure of the Paper

The remainder of the paper is organized as follows. Section 2 reviews the related literature on credit-equity price discovery, CDS indices, regime-switching models, and credit-based hedging. Section 3 describes the data, defines the velocity and volatility concepts that underpin the analysis, and presents the model-identified regimes. Section 4 sets out the econometric methodology, including the Markov switching vector autoregression, its estimation by the Expectation-Maximization algorithm, and the regime-conditional Granger causality framework. Section 5 reports the central results on regime-conditional leadership and their robustness across signal specifications. Section 6 documents the asymmetry between the peak and the trough of a correction, including the confirmed recovery signal and its out-of-sample validation. Section 7 presents the event study of eleven major corrections and the relationship between credit-move magnitude and drawdown depth. Section 8 develops the applications to portfolio and risk management, including the dynamic allocation overlay, drawdown sizing, and credit-based hedging. Section 9 concludes and identifies directions for further research.

2. Literature Review

2.1 Credit Derivatives as Information Vehicles

The theoretical case for credit leading equities rests on informed trading in CDS markets (Acharya and Johnson, 2007), the leverage and directional flexibility of CDS contracts, and the forward-looking default information embedded in spreads. Longstaff, Mithal, and Neis (2005) found information flowing first to CDS and equity markets ahead of bonds. Norden and Weber (2009), by contrast, found equity markets leading both CDS and bond markets, foreshadowing the ambiguity our regime analysis resolves.

2.2 CDS Indices and Regime Dependence

Alexander and Kaeck (2008) showed with a Markov switching model that iTraxx Europe spread determinants are strongly regime-dependent, with equity sensitivity rising sharply in volatile regimes. Fung, Jiang, and Schilling (2008) found CDX volatility leading equity volatility even where spread levels did not, motivating our focus on velocity. These papers anticipate our core theme: the credit-equity relationship is non-linear and state-dependent, and any unconditional test will mislead.

2.3 Price Discovery Between Credit and Equity Markets

A substantial literature investigates where price discovery occurs in the credit-equity system. Blanco, Brennan, and Marsh (2005) showed that for investment-grade firms the CDS market leads the bond market in price discovery, while equity prices in turn lead CDS spreads, establishing an early ordering that our regime analysis qualifies. Forte and Pena (2009) decomposed the dynamic relationships among stock, bond, and CDS markets and found that stocks lead CDS and bonds most of the time, but that leadership reverses in periods of heightened credit risk - an early statement of the regime-dependence we formalize. Hilscher, Pollet, and Wilson (2015) found that equity markets lead credit markets at high frequency for the typical firm, attributing the result to informed traders preferring the more liquid equity venue, yet also documented exceptions concentrated in distressed names. Marsh and Wagner (2016) studied the lead-lag relationship in returns and order flow and concluded that the direction of information flow is state-dependent, varying with the level of market-wide volatility.

The role of CDS markets around news and distress has received particular attention. Acharya and Johnson (2007) found evidence of informed trading in CDS markets ahead of adverse credit news, with the effect increasing in the number of banks lending to a firm, consistent with information leakage from the lending relationship. Ni and Pan (2011) documented predictability from CDS to equity that is strongest when short-sale constraints bind in the equity market, suggesting negative information migrates to CDS when it cannot be expressed in equities. Lee, Naranjo, and Velioglu (2018) showed that CDS-implied credit information predicts equity returns internationally, with

stronger effects during crises. Han and Zhou (2015) extracted a default-risk measure from the term structure of CDS spreads and found it predicts equity returns in the cross-section.

2.4 CDS Indices, Systemic Risk, and Volatility

Research on aggregate CDS indices, rather than single names, is most directly relevant to our use of CDX NA IG. Alexander and Kaeck (2008) modeled iTraxx Europe with a Markov switching framework and found that the determinants of index spreads are strongly regime-dependent, with equity-volatility sensitivity rising sharply in turbulent regimes - the closest methodological precedent for this paper. Fung, Sierra, Yau, and Zhang (2008) examined the relationship between the U.S. equity market and the CDX investment-grade and high-yield indices, finding that the high-yield index is more closely linked to equities and that volatility linkages differ markedly from return linkages. Coudert and Gex (2010) showed that CDS index markets can transmit contagion and that spread changes lead equity volatility during crisis episodes. Tang and Yan (2017) related aggregate CDS spreads to macroeconomic conditions and investor sentiment, finding a strong countercyclical component.

A parallel literature links credit spreads to the macroeconomy and to future real activity. Gilchrist and Zakrajsek (2012) constructed the widely used excess bond premium from corporate credit spreads and showed it has substantial predictive power for economic activity and equity returns, isolating a component of spreads orthogonal to default risk that reflects shifts in risk appetite. Lopez-Salido, Stein, and Zakrajsek (2017) found that elevated credit-market sentiment predicts subsequent reversals in economic growth and risk assets, reinforcing the interpretation of credit conditions as a forward-looking barometer. Giglio, Kelly, and Pruitt (2016) showed that systemic-risk measures, including credit-based ones, have predictive content for the lower tail of the distribution of macroeconomic and market outcomes. Stock and Watson (2003) provide the broader context that asset prices, including credit spreads, are imperfect but useful leading indicators of output and inflation.

2.5 Regime-Switching Models in Asset Markets

Our econometric approach builds on the regime-switching tradition initiated by Hamilton (1989), who introduced the Markov switching autoregression for the business cycle. Ang and Bekaert (2002) demonstrated that international equity correlations are regime-dependent, rising in bear markets, with direct implications for diversification and hedging. Guidolin and Timmermann (2006, 2007) applied multi-state regime-switching models to the joint distribution of stock and bond returns and showed that regime-conditional forecasts improve asset allocation. Krishnan, Ritchken, and Thomson (2010) and Maalaoui Chun, Dionne, and Francois (2014) applied regime-switching specifically to credit spreads, finding that the credit-spread term structure and its determinants shift across latent volatility regimes. These studies collectively motivate allowing the credit-equity relationship to differ across regimes rather than imposing a single linear specification.

2.6 Hedging Equity Risk with Credit Instruments

On the application side, several studies examine credit instruments as equity hedges. Kapadia and Pu (2012) studied the limits of arbitrage between equity and credit markets, finding that the two can decouple precisely when hedging is most desired. Norden and Weber (2009) examined the co-movement of CDS, bond, and stock markets and quantified the conditions under which CDS positions hedge equity exposure. More recent practitioner and academic work has explored credit-index protection and tail-risk hedging overlays, generally concluding that such overlays reduce volatility and drawdown, motivating our focus on the signal's informational content rather than on a standalone trading rule.

2.7 Research Gap

Despite this breadth, the literature leaves four specific gaps that this paper addresses. First, prior studies define stress either by spread level or by exogenous crisis dating; none defines regimes endogenously by the volatility of credit-spread velocity, even though our evidence shows that the turbulence of the rate of change, not the level, is what distinguishes states. Second, prior work documents either unconditional Granger causality or crisis co-movement, but does not establish a regime-conditional leading relationship that is robust across signal specifications. Third, the literature has not isolated the asymmetry between drawdown confirmation at the peak of a correction and a leading recovery signal at the trough. Fourth, recovery signals in prior literature are rarely validated out-of-sample; we show that requiring a confirmed (repeated) tightening sharpens the recovery signal and that the result survives out-of-sample testing on corrections the analysis did not use to define it. This paper fills all four gaps.

3. Data and Regime Definition

3.1 Series

CDX NA IG 5-year on-the-run index spreads (basis points) are obtained from Markit/Bloomberg for September 2011 through June 2026. NASDAQ Composite daily closing levels are from public market data sources. The matched daily sample contains 3,668 observations after computing changes. First differences of CDX and log-returns of NASDAQ are stationary by Augmented Dickey-Fuller tests; all analysis uses these stationary transforms.

A deliberate feature of our sample is that it begins in September 2011, well after the 2007–2009 Global Financial Crisis, and therefore covers the CDX market in its modern, post-reform form. This matters because the structure of the credit-derivatives market changed profoundly after the crisis, and pre-crisis CDS dynamics are not representative of how the market trades today. The gross notional outstanding of CDS collapsed from roughly $61–62 trillion at end-2007 to about $26 trillion by 2010 and to under $10 trillion by 2017, as participants compressed redundant contracts and reduced exposures. Just as importantly, post-crisis regulation, the Dodd-Frank Act in the United States and EMIR in Europe, mandated central clearing through central counterparties, trade reporting, and margin requirements for standardized index products such as CDX NA IG (restrictions on uncovered, or "naked," positions applied more narrowly, notably to European sovereign CDS rather than to broad corporate indices). By 2022 the large majority of CDS notional was executed on swap-execution facilities and cleared centrally. Index products such as CDX NA IG, being the most standardized instruments, were precisely those most amenable to clearing and now dominate trading activity. Because these reforms increased transparency and reduced the informational opacity that once made CDS a venue for informed trading, the credit-equity relationship we measure reflects the contemporary, regulated, centrally cleared market rather than the opaque pre-crisis one.

3.2 Velocity and Volatility: Definitions

Because the distinction is central to this study, we define two related but conceptually distinct quantities precisely. We use them consistently throughout, and the reader should keep them separate.

Velocity denotes the signed rate of change of the CDX spread over a fixed horizon. In our canonical specification this is the three-day percentage change, $(S_t - S_{t-3})/S_{t-3}$. Velocity is a property of a single observation: it has both a direction (positive denotes widening, negative denotes tightening) and a magnitude, and it answers the question of how fast, and in which direction, the spread is moving at a given moment. A widening signal when the three-day change exceeds +3%, and a tightening signal when it falls below −5% are directional velocity events in this paper.

Volatility denotes the dispersion of those velocity readings: the standard deviation (σ) of the velocity series, estimated over a period of time. Volatility is a property of a period rather than of a single day, it is non-directional and always non-negative, and it answers the question of how turbulent the spread has been. Formally, volatility is the standard deviation of the velocity series; velocity is the raw directional signal, and volatility is the statistical summary of how scattered those signals are.

The architecture of our model rests on this separation. Regimes are defined by volatility: the Markov switching estimation sorts days into Low-, Moderate-, and High-Stress clusters according to the standard deviation of the three-day velocity, so that a regime characterizes the turbulence of an environment over a period. Signals, by contrast, are velocity readings: the widening and tightening triggers are single-day directional events evaluated within a given regime. The two operate at different levels: volatility identifies the environment, while velocity is the directional signal acted upon within it. The paper's key result, the high-stress recovery signal, combines them: a tightening velocity occurring within the high-volatility regime. Crucially, both differ from the spread level: as Table 1 shows, the Moderate- and High-Stress regimes sit at almost identical average levels (about 71 and 70 bps), so neither velocity nor volatility is captured by the level, and only a volatility-based regime definition distinguishes the two.

3.3 Regime Identification

We define the Markov-Switching Vector Autoregression (MS-VAR) state variable as the 3-day percentage change in CDX spreads, paired with the daily NASDAQ return. The Expectation-Maximization (EM) algorithm sorts observations into three regimes by the volatility of this velocity measure (the standard deviation of the 3-day percentage change), then labels them by ascending volatility. The resulting regimes are summarized in Table 1. Critically, the regimes are not separated by spread level: the weighted-median CDX level is similar across the Moderate and High regimes (about 71 and 70 bps respectively), confirming that the model classifies stress by velocity, not level.

Table 1: Model-Defined Volatility Regimes (state = 3-day % change, i.e. velocity, of CDX NA IG)

Metric	Low-Stress	Moderate-Stress	High-Stress
Share of days	63.1%	29.4%	7.5%
σ (3-day % change)	2.22%	2.99%	**7.61%**
Median |3-day % change|	2.18%	3.13%	8.80%
Weighted-median CDX level	~63 bps	~71 bps	~70 bps
ρ(CDX Δ, NASDAQ ret)	−0.74	−0.82	−0.67

The near-identical spread levels of the Moderate and High regimes, despite a 2.5-fold difference in the volatility of velocity, is the central justification for the velocity-based approach. A level-threshold model would assign both regimes to the same bucket and miss the distinct dynamics documented below.

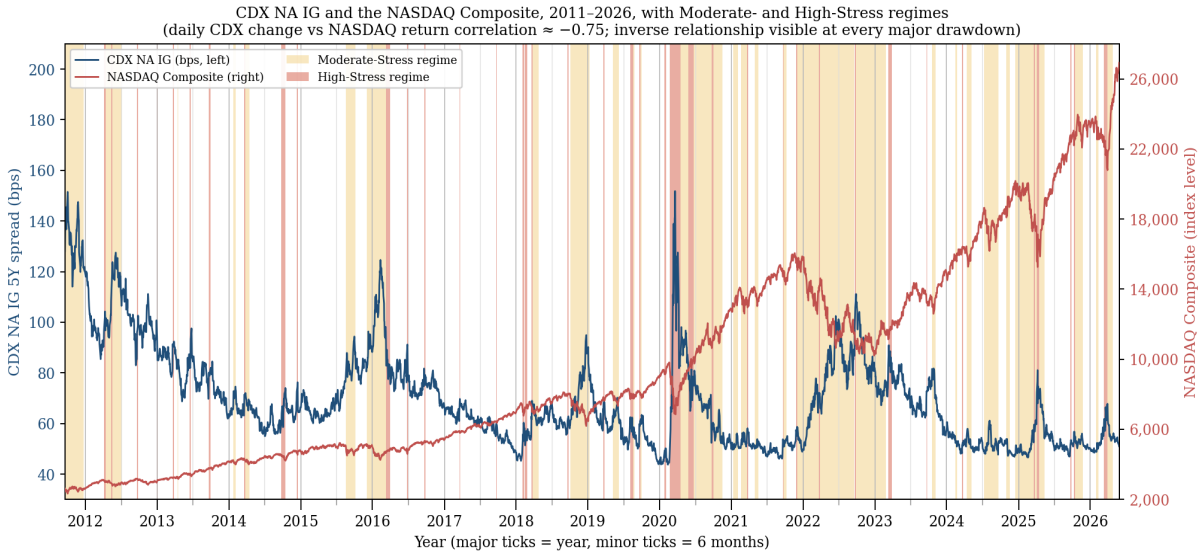


Figure 1: CDX NA IG 5-year spread (left axis) and the NASDAQ Composite (right axis), September 2011 to June 2026. Shaded bands mark the model-identified Moderate-Stress (amber) and High-Stress (red) regimes; the Low-Stress regime is left unshaded. The strong negative co-movement between the two series is visible at every major drawdown, most dramatically in March 2020, when the CDX spread spiked above 150 bps as the NASDAQ fell sharply.

Figure 1 illustrates the two central features that motivate the analysis. First, the CDX spread and the NASDAQ move inversely: every pronounced spike in the credit spread coincides with a decline in the equity index, consistent with the strong negative daily correlation of about −0.75. Second, the shaded Moderate- and High-Stress regimes cluster precisely around the credit-spread peaks and the associated equity drawdowns, while the prolonged unshaded stretches correspond to calm, rising markets, providing visual confirmation that the model isolates economically meaningful episodes of credit stress.

4. Methodology

4.1 The Vector Autoregression (VAR)

We begin with a vector autoregression, a standard model in which each variable is explained by its own recent past and by the recent past of the other variable. With two variables - the change in CDX spreads and the NASDAQ return - and a lag length of three days, the VAR asks a simple question: once we already know the last three days of NASDAQ returns, do the last three days of CDX changes add any extra ability to explain today's NASDAQ return? If they do, CDX is said to carry information about the future of NASDAQ.

4.2 Allowing the Relationship to Switch: the Markov-Switching VAR

A single VAR forces one fixed relationship across all market conditions. That is unrealistic, because the link between credit and equities is plainly different in a crash than in a calm market. We therefore use a Markov-Switching VAR (MS-VAR), which allows the model to operate in three distinct "regimes," each with its own coefficients and its own volatility:

$$y_t = c(s_t) + A_1(s_t)\, y_{t-1} + A_2(s_t)\, y_{t-2} + A_3(s_t)\, y_{t-3} + \varepsilon_t, \quad \text{with regime } s_t \in \{\text{Low, Moderate, High}\}$$

Here y_t stacks the two variables, and the term s_t denotes which of the three regimes the market is in on day t. The regime is not observed directly; it is inferred from the data. Crucially, the regime is allowed to change from day to day according to a Markov chain, meaning the probability of being in each regime tomorrow depends only on which regime we are in today, summarized by a 3×3 matrix of transition probabilities.

4.2.1 How the model is estimated: the EM algorithm

Because we never directly observe which regime the market is in on a given day, we cannot simply run a regression. Instead we use the Expectation-Maximization (EM) algorithm, an iterative procedure that alternates between two steps until the answers stop changing:

In the E-step (Expectation), the model uses its current best guess of the parameters to compute, for every day, the probability that the day belonged to each of the three regimes. These probabilities are produced by a standard filtering recursion (the Hamilton filter) that sweeps forward through the data and a complementary backward sweep.

In the M-step (Maximization), the model takes those regime probabilities as given and re-estimates the parameters - the VAR coefficients and the volatility of each regime - by weighted least squares, where each day is weighted by how likely it is to belong to the regime in question. The two steps repeat: better parameters produce sharper regime probabilities, which in turn produce better parameters. We stop when successive iterations change the parameters by a negligible amount (less than one part in a million), which typically occurs within roughly eighty iterations. The three regimes are then labelled Low, Moderate, and High by ascending credit volatility.

4.3 Testing for Leadership: Granger Causality and the F-statistic

To ask whether CDX leads NASDAQ within a given regime, we use a Granger causality test. The idea is to compare two nested models of the NASDAQ return: a restricted model that uses only past NASDAQ returns, and an unrestricted model that adds past CDX changes. If adding the CDX terms produces a statistically meaningful improvement in fit, then CDX is said to Granger-cause (lead) NASDAQ.

The improvement is measured by an F-statistic, which compares the reduction in unexplained variation against what would be expected by chance given the number of extra terms added. A larger F-statistic indicates a stronger relationship. We convert each F-statistic into a p-value, the probability of observing a relationship that strong if there were truly none. By convention a relationship is called statistically significant when the p-value falls below 0.05, meaning less than a one-in-twenty chance of arising by luck; we mark $p < 0.05$ with one asterisk, $p < 0.01$ with two, and $p < 0.001$ with three. We run the test in both directions - CDX→NASDAQ and NASDAQ→CDX - weighting each day by its regime probability, so that each regime is assessed on the days that genuinely belong to it.

4.4 Signal Definitions

Two operational signals are derived from CDX velocity: a widening signal (a three-day change above +3%) and a tightening signal (a three-day change below −5%). These are chosen, economically motivated cutoffs, not model outputs: the regimes are model-derived, but the signal levels are set by judgment and tested for robustness. Each level is calibrated against the typical move-size observed in each regime (Table 3a):

- The +3% widening level matches the typical three-day move-size of the Moderate-Stress regime (median 3.1%, versus 2.2% in Low-Stress), so a move of this size is characteristic of conditions that have left calm markets, even though the regime itself is defined by the volatility of velocity rather than by any single move.
- The −5% tightening level is larger in magnitude, falling between the typical Moderate- and High-Stress moves (3.1% and 8.8%). This reflects the empirical asymmetry that recoveries are marked by sharper spread compressions than the moves accompanying widening.

In each case a round figure near the center of the plausible range is used, not the value that maximizes any result. The recovery finding does not hinge on this choice: a sensitivity analysis (Section 6.3, Table 5c) shows the signal is smooth and monotonic across thresholds from −3% to −8%, confirming −5% is representative rather than fitted.

For each signal we evaluate concurrent NASDAQ behavior, forward returns at multiple horizons, and behavior across the eleven corrections in the event study. The tightening signal is further refined into a confirmed signal, a second tightening within roughly one month, which Section 6 shows materially improves its reliability.

5. Results: Regime-Conditional Leadership

5.1 The Core Finding

Table 2 reports regime-conditional Granger causality for the canonical specification (3-day percentage change). Two questions are distinct and both matter: (i) does each direction clear the bar of statistical significance, and (ii) how the two directions compare in strength. The headline result concerns the first question. In the Low-Stress regime, CDX does not significantly help predict NASDAQ (F = 2.13, p = 0.095, above the 0.05 threshold). But in both the Moderate- and High-Stress regimes, CDX crosses into clear significance as a predictor of NASDAQ (F = 5.19, p = 0.001 and F = 4.99, p = 0.002). In other words, CDX acquires genuine predictive information about NASDAQ precisely when credit stress is elevated.

Table 2: Regime-Conditional Granger Causality (3-day % change specification)

Regime	CDX→NDQ F	p-value	NDQ→CDX F	CDX as predictor
Low-Stress	2.13	0.095	142.4	No (p>0.05)
Moderate-Stress	**5.19**	0.001	96.9	**Yes ****
High-Stress	**4.99**	0.002	11.7	**Yes ****

A necessary clarification on the two F-statistics. We note that the NASDAQ→CDX F-statistic is numerically larger than the CDX→NASDAQ F-statistic in every regime. This does not contradict the finding. The two markets are tightly coupled, and because both respond to common shocks the backward direction (NASDAQ→CDX) is mechanically strong everywhere. The economically interesting and non-trivial fact is the behavior of the forward direction: whether CDX carries any genuine predictive information about NASDAQ at all. That forward channel is absent in calm markets (not significant) and switches on in elevated stress (significant at the one-percent level). The contribution of this paper is therefore precise: we do not claim that CDX dominates NASDAQ in stress, but that CDX becomes a statistically significant predictor of NASDAQ specifically in elevated-stress regimes, having no such power in calm conditions. This regime-dependent activation of the credit-to-equity channel is exactly the pattern anticipated by Forte and Pena (2009) and Marsh and Wagner (2016), and it reconciles the average-case equity leadership documented by Hilscher, Pollet, and Wilson (2015) with the distress-period exceptions they also observed.

5.2 Robustness Across Specifications

To ensure the regime-conditional finding is not an artifact of the chosen signal definition, we re-estimate the model under six specifications: 1-, 3-, and 5-day horizons, each in basis points and in percentage change. Because each specification defines its regimes by the velocity of CDX moves, the magnitude of move that separates one regime from the next differs by specification. Table 3a reports the typical (median) absolute move in each regime, so the F-statistics in Table 3b can be

read in context: for example, in the canonical 3-day percentage specification a typical Moderate-Stress day involves about a 3% three-day move and a typical High-Stress day about a 9% move, versus roughly 2% in calm markets. Table 4 then summarizes the leadership verdict.

Table 3a: Typical Move-Size in Each Regime (median absolute CDX move)

Specification	Low-Stress	Moderate-Stress	High-Stress
1-day Δ (bps)	0.7 bps	1.7 bps	5.0 bps
3-day Δ (bps)	1.2 bps	2.8 bps	8.1 bps
5-day Δ (bps)	1.5 bps	4.0 bps	10.5 bps
1-day % change	1.3%	1.8%	6.2%
3-day % change	2.2%	3.1%	8.8%
5-day % change	2.6%	4.6%	11.5%

Table 3b: CDX→NASDAQ Granger F-statistic by Specification and Regime

Specification	Low-Stress	Moderate-Stress	High-Stress
1-day Δ (bps)	2.28	4.88 **	3.37 *
3-day Δ (bps)	1.00	5.01 **	3.71 *
5-day Δ (bps)	2.25	2.93 *	1.18
1-day % change	2.91 *	5.17 **	3.90 **
3-day % change	2.13	5.19 **	4.99 **
5-day % change	2.29	2.92 *	1.61

Table 4: Leadership Verdict — Is CDX→NASDAQ Significant (p<0.05)?

Specification	Low-Stress	Moderate-Stress	High-Stress
1-day Δ (bps)	no	**CDX leads**	**CDX leads**
3-day Δ (bps)	no	**CDX leads**	**CDX leads**
5-day Δ (bps)	no	**CDX leads**	no
1-day % change	(marginal)	**CDX leads**	**CDX leads**
3-day % change	no	**CDX leads**	**CDX leads**
5-day % change	no	**CDX leads**	no

Summary: CDX→NASDAQ is significant in Low-Stress in 1 of 6 specifications, in Moderate-Stress in 6 of 6, and in High-Stress in 4 of 6. The Moderate-Stress leadership is therefore exceptionally robust, holding under every specification. High-Stress leadership holds under all short-horizon (1- and 3-day) specifications and fails only under the 5-day window, which is too slow to capture the fast, self-reversing moves that characterize acute stress. Calm-market leadership is essentially absent. The regime-conditional finding is a structural feature of the data, not a specification artifact.

These results align with prior evidence that the credit-equity relationship is state-dependent. Alexander and Kaeck (2008) found, in a Markov switching framework, that the sensitivity of CDS index spreads to equity-market factors rises sharply in volatile regimes, and Fung, Sierra, Yau, and Zhang (2008) reported that the link between CDX and equities strengthens in turbulent periods. Our finding that the credit-to-equity channel attains significance specifically in the moderate- and high-stress regimes is consistent with both, and extends them by showing the result is robust across every signal specification tested.

6. The Asymmetry Across a Correction: Confirmation at the Peak, Leadership at the Trough

6.1 Widening Confirms the Decline; Tightening Signals the Turn

The relationship between CDX and NASDAQ is strongest contemporaneously, with a same-day correlation of −0.74 to −0.82 across regimes and near-zero one-day-ahead correlation. A sharp widening of CDX (a three-day change above +3%) is therefore primarily a coincident confirmation: when it fires, the NASDAQ is already declining over the same window in 85 to 91 percent of cases, so widening reliably signals that a drawdown is genuinely underway. Its forward content is conditional rather than universal. In calm markets a widening signal is largely a transient blip that mean-reverts, but when it occurs in the Moderate- or High-Stress regime, or accompanies a genuine macro-financial event, the decline tends to continue: the event study in Section 7 shows that in such episodes the NASDAQ trough still lies a median of 28 trading days ahead of the widening signal (Table 6a). In practice, then, a widening signal is best read together with the prevailing regime and the news flow: the same +3% move is noise in a calm tape but an early warning of a deeper drawdown when a real stress event is developing.

6.2 The Tightening Recovery Signal Is Concentrated in High Stress

A sharp tightening of CDX (a three-day change below −5%) is a genuinely leading signal of the recovery, and its predictive power is concentrated almost entirely in the High-Stress regime. Within high stress, a tightening is followed by significantly positive NASDAQ returns over the following 10 to 20 trading days, whereas in calm or moderate conditions the same move carries no forward information. The strength of this signal, and the gain from requiring confirmation, is quantified in Section 6.3.

The asymmetry is therefore directional as well as regime-dependent: into a drawdown, credit and equities sell off together and widening is coincident; emerging from a drawdown, credit leads, and tightening within high stress anticipates the equity recovery. March 2020 is the archetype, when the CDX spread peaked on the very day of the NASDAQ low and then compressed sharply as policy support arrived.

The mirror-image question is how often a tightening signal is followed by a genuine recovery rather than a continued fall. Table 5a reports this for the confirmed tightening (defined in Section 6.3) across the full sample. Twenty trading days after a confirmed tightening the NASDAQ is higher 75.5% of the time, and across 99 distinct episodes it is higher in 68%; the complementary cases, in which the market kept falling, occur about a quarter of the time. The signal therefore carries a real and quantifiable edge while remaining probabilistic.

Table 5a: Accuracy of the Confirmed Tightening Signal (2nd 3-day move < −5% within 20 days), full sample 2011–2026

Outcome measure	Recovery (correct)	Kept falling (false)	Basis
NASDAQ higher 5 trading days later	63.8%	36.2%	359 signal-days
NASDAQ higher 10 trading days later	73.8%	26.2%	359 signal-days
NASDAQ higher 20 trading days later	75.5%	24.5%	359 signal-days
NASDAQ higher 20 days later (distinct episodes)	68%	32%	99 episodes

Note: "Recovery" means the NASDAQ was higher at the stated horizon; "kept falling" means lower. Daily signal-days overstate the count owing to within-episode correlation, so the episode row de-clusters signals more than ten trading days apart.

This directional asymmetry is consistent with the price-discovery literature. Forte and Pena (2009) and Marsh and Wagner (2016) document that the direction of information flow between credit and equity markets shifts with the level of stress, while Acharya and Johnson (2007) show that informed credit trading is concentrated in adverse states. Our result refines this body of work by locating the credit market's forward-looking advantage specifically at the recovery: the tightening of spreads that follows acute stress leads the equity rebound, consistent with risk appetite returning to the more liquid credit market first.

6.3 A Confirmed (Repeated) Tightening Sharpens the Signal

Because a single sharp tightening can be a transient rebound within an ongoing decline, we ask whether requiring confirmation improves reliability. We classify each tightening day (three-day change below −5%) as confirmed when at least one other tightening day occurred within the prior 20 trading days, and as single otherwise. A second tightening suggests the credit market is repricing risk downward in a sustained way rather than reacting momentarily.

The refinement is strongly supported. Forward NASDAQ returns after a confirmed tightening average +2.43% over the following 20 trading days and are positive 75% of the time, compared with +0.72% (positive 64%) after a lone, unconfirmed move; the difference is statistically significant (Welch $t = 2.50$, $p = 0.014$). Because signal-days within an episode are correlated, we also test at the episode level: across 99 distinct confirmed episodes the mean 20-day return is +1.62% (one-sample $t = 3.10$, $p = 0.003$), with a block-bootstrap 95% confidence interval of [+0.62%, +2.65%] that excludes zero. Requiring a second tightening thus filters out isolated rebounds and converts a weak signal into a meaningful one. Table 5b summarizes the comparison.

Table 5b: Single versus Confirmed (Repeated) Tightening Signal — Forward NASDAQ Returns

Signal type	n	Fwd 10d	Fwd 20d	P(up) 20d
Single tightening (1st in 20d window)	61	−0.19%	+0.72%	64%
Confirmed tightening (≥2 in 20d window)	**359**	**+1.28%**	**+2.43%**	**75%**

Note: P(up) is the share of signals after which the NASDAQ was higher at the end of the forward window, i.e. the empirical probability of a positive return.

Because the regimes are derived from the Markov switching model but the signal thresholds are chosen, economically motivated cutoffs rather than optimized parameters, we confirm that the

recovery result does not depend on the specific choice of −5%. Table 5c varies the tightening threshold from −3% to −8%. The edge is smooth and monotonic: as the threshold is made stricter, the average recovery rises while the sample shrinks, so significance weakens through loss of data rather than disappearance of the effect. The −5% cutoff is a representative round number in the middle of this stable range and is deliberately not the value that maximizes the result. The recovery signal is therefore a robust feature of the data, not an artifact of a finely tuned threshold.

Table 5c: Sensitivity of the Confirmed Recovery Signal to the Tightening Threshold (forward 20-day NASDAQ return, full sample)

Tightening threshold	n (confirmed)	Mean fwd 20d	P(up) 20d	vs baseline p
−3%	844	+1.75%	72%	0.087
−4%	572	+2.04%	74%	0.008
−5%	**359**	**+2.43%**	**75%**	**0.0009**
−6%	233	+2.86%	78%	0.0001
−7%	132	+2.85%	73%	0.007
−8%	84	+3.20%	74%	0.011

Note: each row applies the confirmed-tightening rule at the stated threshold. The −5% row (shaded) is the specification used throughout. The effect is present and economically larger at stricter thresholds; rising p-values at −7% and −8% reflect the shrinking sample, not a weaker effect.

6.4 Out-of-Sample Validation with Real-Time Regime Probabilities

A signal measured on the same data used to discover it may fit the idiosyncrasies of that history rather than a durable relationship. We therefore validate the confirmed recovery signal out-of-sample. The sample is split into an in-sample period, on which the relationship is observed, and an out-of-sample period set aside and treated as the unknown future; if the signal continues to work on data it never informed, that is evidence the pattern is genuine.

We place the split at the start of 2023. This is deliberate: the 2020 COVID crash is the single most extreme and most favorable episode for a credit-based recovery signal, since the CDX spread peaked on the exact day of the NASDAQ trough, so if it fell in the out-of-sample window it could flatter the result. Holding it (and the 2022 bear market) in the in-sample period leaves an out-of-sample window of ordinary, moderate corrections — the 2023 banking stress, the 2024 yen-carry unwind, the 2025 tariff shock, and the 2026 Middle East escalation — a more demanding and more honest test.

The confirmed recovery signal passes convincingly. On the out-of-sample data the confirmed tightening was followed by an average NASDAQ gain of 5.01% over the subsequent 20 trading days, positive on 85% of occasions, against an in-sample figure of 1.90%. Tested at the episode level, the 17 independent out-of-sample episodes have a mean 20-day return of +3.09% with a block-bootstrap 95% confidence interval of [+1.03%, +5.08%] ($p = 0.001$), so the result is not driven by correlated signal-days. The effect is not weaker out of sample; if anything it is stronger in the recent period of moderate corrections, which indicates the signal is not an artifact of the COVID episode. Table 5d summarizes the in-sample and out-of-sample comparison.

As a still stricter check that avoids look-ahead in the regime assignment itself, we re-estimate the Markov switching model on an expanding window and classify each day using only filtered (real-time) regime probabilities, available from information up to that day. Conditioning the confirmed tightening on a real-time filtered High-Stress probability above one-half isolates nine genuine high-stress recovery signals across the out-of-sample period, every one of which was followed by a higher NASDAQ twenty days later (mean +6.36%). The recovery signal therefore survives both an out-of-sample split and a fully real-time regime classification.

Table 5d: Out-of-Sample Validation of the Confirmed Tightening Signal (in-sample through 2022; out-of-sample 2023–2026)

Period / signal	n	Fwd 20d	P(up) 20d	vs base p
In-sample (≤2022), confirmed	298	+1.90%	73%	—
In-sample (≤2022), single	48	+0.62%	65%	—
Out-of-sample (2023–26), confirmed	**61**	**+5.01%**	**85%**	**<0.001**
Out-of-sample (2023–26), single	13	+1.10%	62%	n.s.

Note: the out-of-sample significance is computed at the episode level using a block bootstrap (17 independent confirmed episodes), not from correlated signal-days. The filtered real-time variant described in the text isolates nine high-stress recovery signals, all positive at 20 days.

7. Event Study: Eleven Major NASDAQ Corrections

We summarize the eleven corrections through two complementary lenses. Table 6a tracks the widening signal, which confirms a decline is underway: for each event it reports the first widening signal (three-day CDX change above +3%), how many trading days that signal followed the NASDAQ peak, how many remained to the trough, the peak-to-trough drawdown duration, the CDX magnitude, the NASDAQ decline still to come from the signal date to the trough, and the full drawdown depth.

Table 6a: The Widening Signal Across Eleven Corrections — Timing Relative to the Peak and Trough, Drawdown Duration, CDX Magnitude, and Drawdown Depth (2011–2026)

Event	Scope	Widening signal	Days after peak	Days to trough	NDQ DD days	CDX peak (bps)	CDX move %	NDQ DD signal→trough	NDQ DD peak→trough
2012-06 EU periphery	Narrow	2012-04-05	8	39	47	128	+31%	−10.8%	−10.5%
2015-08 China selloff	Narrow	2015-07-21	1	25	26	88	+24%	−13.5%	−12.9%
2016-02 Oil/credit rout	Sectoral	2015-12-15	10	39	49	125	+41%	−14.6%	−16.5%
2018-12 Q4 crash	Global	2018-10-19	14	43	57	95	+37%	−16.9%	−18.2%
2020-03 COVID crash	Global	2020-02-21	2	21	23	152	**+203%**	−28.4%	−30.1%
2022-02 Russia/Ukraine war	Narrow	2022-01-18	15	71	86	78	+29%	−15.0%	−12.7%
2022-06 Rate-hike peak	Global	2022-05-06	27	28	55	102	+17%	−12.3%	−17.9%
2023-03 SVB / banking	Sectoral	2023-02-21	12	13	25	91	+24%	−3.1%	−8.0%
2024-08 Yen-carry unwind	Global	2024-07-18	6	14	20	61	+26%	−9.4%	−13.1%
2025-04 Tariff shock	Global	2025-03-20	21	13	34	81	+46%	−13.7%	−17.7%
2026-03 Middle East escalation	Narrow	2026-02-17	13	29	42	68	+36%	−7.9%	−12.8%

Note: "Days after peak" is the trading days from the NASDAQ peak to the first widening signal; "Days to trough" is from that signal to the trough; "DD days" is the peak-to-trough duration; "NDQ DD signal→trough" is the NASDAQ decline from the widening-signal date to the trough (the downside remaining after the signal fires, a median of about −13%); "NASDAQ drawdown" is the full peak-to-trough decline. The widening signal lags the peak by a median of 12 trading days and precedes the trough by a median of 28. Scope shading: green = Narrow, amber = Sectoral, red = Global.

Table 6b turns to the other side of the correction, the confirmed tightening that marks the turn. For each event it reports the confirmed tightening nearest the NASDAQ trough (a second three-day move below −5% within roughly one month) and its distance from the trough. Whereas the widening signal precedes the trough by a median of 28 trading days, the confirmed tightening sits within a median of just three trading days of it.

Table 6b: The Confirmed Tightening Signal Across Eleven Corrections — Proximity to the NASDAQ Trough (nearest-trough rule)

Event	Scope	NDQ trough	Confirmed tightening signal	Days: signal to trough	NDQ +20d
2012-06 EU periphery	Narrow	2012-06-01	2012-07-02	**21 after**	−0.4%
2015-08 China selloff	Narrow	2015-08-25	2015-08-28	**3 after**	−5.9%
2016-02 Oil/credit rout	Sectoral	2016-02-11	2016-02-17	**4 after**	+5.1%
2018-12 Q4 crash	Global	2018-12-24	2018-12-28	**2 after**	+6.7%
2020-03 COVID crash	Global	2020-03-23	2020-03-23	**at trough**	+20.4%
2022-02 Russia/Ukraine war	Narrow	2022-04-29	2022-03-30	**21 before**	−10.9%
2022-06 Rate-hike peak	Global	2022-06-16	2022-05-27	**13 before**	−7.8%
2023-03 SVB / banking	Sectoral	2023-03-10	2023-03-06	**4 before**	+4.4%
2024-08 Yen-carry unwind	Global	2024-08-07	2024-08-16	**7 after**	−0.2%
2025-04 Tariff shock	Global	2025-04-08	2025-04-09	**1 after**	+4.7%
2026-03 Middle East escalation	Narrow	2026-03-30	2026-04-01	**2 after**	+14.0%

Note: "Confirmed signal" is the confirmed tightening nearest the trough; "Days: signal to trough" is the number of trading days between that signal and the trough ("before" = preceded the trough, "after" = followed it); "NDQ +20d"

is the NASDAQ return over the 20 trading days after the signal. The confirmed tightening landed within a week of the trough in eight of eleven episodes. Under this nearest-trough rule the NASDAQ was higher 20 days later in six episodes; the cases where it preceded the trough by about three weeks (2022 Russia-Ukraine and rate-hike) saw a second selling leg, consistent with the probabilistic nature of the signal.

Across the eleven corrections the median drawdown was −13.1% and the median peak-to-trough duration was 42 trading days. Tables 6a and 6b separate the two CDX signals along the way: the widening signal that confirms a drawdown is in progress (Table 6a), and the confirmed tightening signal that marks its turn (Table 6b). Two facts about their timing matter for interpretation.

First, the widening signal confirms rather than predicts. In every episode the NASDAQ peak precedes the first widening signal by one to two weeks (a median of 12 trading days), so the widening signal does not warn of a correction from a calm market; it corroborates a decline already under way, while a substantial portion of the drawdown still lies ahead (a median of 28 further trading days to the trough).

Second, the confirmed tightening clusters tightly around the trough. The confirmed tightening nearest the low fell within a median of about three trading days of the NASDAQ trough, and within a week of it in eight of the eleven episodes — a far closer marker of the turn than the widening signal provides. Under this nearest-trough rule the NASDAQ was higher twenty days later in six of the eleven episodes; the two cases in which the nearest confirmed tightening preceded the trough by about three weeks (the 2022 Russia-Ukraine shock and the 2022 rate-hike bear market) were episodes with a second leg of selling, consistent with the probabilistic nature of the signal established in Section 6. The episode-level and out-of-sample evidence in Section 6 quantifies this reliability across the full sample rather than these eleven episodes alone.

7.1 CDX Move Magnitude as a Measure of Drawdown Depth

Tables 6a and 6b also expose a second, orthogonal dimension of information in CDX beyond signal timing: the magnitude of the credit move is a strong contemporaneous gauge of how deep the NASDAQ drawdown will be. Across the eleven episodes, the percentage size of the CDX move is highly correlated with the eventual drawdown (Pearson $r = -0.85$, $p < 0.001$), as are the absolute basis-point move ($r = -0.84$) and the peak three-day velocity ($r = -0.80$). The larger the credit dislocation, the deeper the equity decline. The clearest illustration is the COVID crash, whose 203% CDX explosion accompanied the deepest drawdown in the sample (−30.1%), while milder credit moves such as the 2024 yen-carry unwind (+26%) accompanied shallower declines (−13.1%).

Event scope reinforces the same ordering. Global or systemic events (the 2018 global tightening, COVID, the 2022 rate cycle, the yen-carry unwind, and the 2025 tariff shock) produced a mean drawdown of −19.4%, versus −12.2% for the narrow regional and sectoral events, a gap of about seven percentage points (Mann-Whitney $p = 0.004$). However, the breadth label and the CDX move size carry overlapping information: systemic shocks are precisely those that transmit through

the global funding and credit system, and the CDX move already encodes that severity. In our sample the CDX move magnitude tracks drawdown depth more reliably than the scope label alone, so the two are best read together, with magnitude as the continuous, quantitative measure and scope as the qualitative context. The SVB episode is the instructive edge case: a sharp credit event by velocity, but contained to U.S. regional banks by rapid policy intervention, it produced the shallowest drawdown in the sample (−8.0%).

A crucial caveat governs how this relationship can be used. The magnitude–depth correlation is measured at the CDX peak; that is, using the full move after it has occurred. At the moment the widening signal first fires, one cannot know whether the move will stop at a few percent or escalate to the triple-digit dislocation seen in 2020. The relationship therefore allows an analyst to read severity off CDX as the move develops, updating the expected downside as the credit dislocation grows, but it does not forecast the ultimate size of the CDX move from the initial signal. Neither does the event-scope classification, which is itself a hindsight judgment in the early days of a shock. CDX is thus a real-time severity gauge to be read as it evolves, not a predictor of its own future range. Read together, the two dimensions of CDX, namely the volatility of velocity (which identifies the regime and the timing) and the magnitude of the move (which scales the expected depth), give a fuller picture of an unfolding correction than either alone.

8. Applications in Portfolio and Risk Management

The findings translate into concrete practice for an investor or risk manager who holds NASDAQ exposure and wishes to manage it through the cycle. We organize the applications around the three roles CDX plays (coincident gauge, magnitude-scaled trimming signal, and high-stress recovery signal) and then turn to broader uses in risk systems.

The applications below are consistent with established findings on credit-based hedging. Norden and Weber (2009) quantify the conditions under which CDS positions hedge equity exposure, and Kapadia and Pu (2012) show that equity and credit markets can decouple, which bounds the reliability of any single-instrument hedge. Our overlay reflects both insights: it uses CDX where the credit-equity linkage is strongest (in elevated-stress regimes) and treats the signal as a risk-management tool rather than a guaranteed hedge.

8.1 Dynamic Equity Allocation Overlay

The central application is a rules-based overlay on a core NASDAQ position. In calm markets, where CDX carries no predictive information, the investor remains fully allocated and ignores minor credit fluctuations, avoiding the whipsaw an over-sensitive signal would cause. When CDX widens sharply, in a move the model associates with entry into the moderate- or high-stress regime, the overlay calls for trimming equity exposure, with the size of the trim scaled to the magnitude of the credit move, since CDX magnitude is strongly correlated (about −0.85) with the eventual drawdown depth. A modest widening warrants a modest reduction; a violent dislocation of the kind seen in March 2020 warrants a far larger one. When CDX subsequently tightens sharply from within the high-stress regime, the overlay signals that the correction low is likely in and that the trimmed exposure should be rebuilt. The result is a position that is systematically lighter through the worst of a drawdown and re-risked near the bottom, without requiring any forecast of corrections from calm markets.

8.2 Risk Monitoring and Early Confirmation

Independent of any trading rule, CDX serves as a high-confidence, real-time confirmation tool within a risk-monitoring framework. Because a sharp CDX move coincides with NASDAQ declines in roughly 85 to 91 percent of cases, a risk desk can treat a sudden widening as corroborating evidence that an emerging equity sell-off is a genuine stress event rather than noise, distinguishing a true risk-off episode from an ordinary pullback. This is particularly valuable for cross-asset desks, where a single, liquid, continuously priced credit index summarizes systemic stress more cleanly than a basket of equity indicators alone.

8.3 Drawdown Estimation and Stress Sizing

The magnitude–depth relationship gives risk managers a live, market-based estimate of how severe a developing drawdown may become. As a credit dislocation unfolds, the size of the CDX move provides a continuously updated read on likely equity downside, which can feed directly into

dynamic stop-loss levels, value-at-risk overlays, and scenario analysis. Rather than relying solely on historical-distribution assumptions, a desk can anchor its downside estimate to the severity the credit market is pricing in real time with the caveat, established in Section 7, that the ultimate size of the CDX move cannot be known at the first signal and the estimate must be revised as the move develops.

8.4 Hedging and Tail-Risk Management

For mandates permitted to use derivatives, the same signals inform a credit-based hedging program. Long CDX protection appreciates as spreads widen and is itself a direct equity-correlated hedge, given the strong negative correlation between CDX changes and NASDAQ returns. The regime framework helps time and size such hedges: protection is most justified once the moderate- or high-stress regime is entered, and can be unwound when the high-stress tightening signal indicates the stress has passed. Because index CDS are liquid and centrally cleared in the modern market, this is operationally practical at institutional scale. We emphasize that this is a risk-management application that reduces the volatility and depth of drawdowns, rather than a source of excess return; the evidence does not support credit-based equity timing as a way to beat a buy-and-hold position over a full cycle.

8.5 Practical Considerations and Limitations

Several practical points temper these applications. Transaction costs and the bid-offer on equity and CDS positions argue for partial, scaled adjustments rather than wholesale moves, consistent with the overlay design above. The strongest recovery evidence rests on a limited number of genuine high-stress episodes, so position sizing should be conservative and the signals treated as one input among several rather than a mechanical trigger. Judging in real time whether an unfolding shock is regional or systemic remains difficult, although CDX magnitude helps quantify severity as events develop. Finally, all of these applications are calibrated on the post-crisis, centrally cleared CDX market; should the market structure change materially again, the relationships would need to be re-estimated. None of the foregoing constitutes investment advice; it describes how the statistical findings could inform a disciplined risk-management process.

9. Conclusion

We began by framing CDX NA IG not as a device for forecasting corrections from tranquil conditions but as an overlay to help a long-horizon NASDAQ investor size equity exposure through stress. Read against that goal, the evidence is encouraging and, importantly, it sorts cleanly by the stage of a correction. The practical implication is a single coherent playbook for adjusting allocation rather than a collection of disconnected statistics.

In calm markets, CDX carries no predictive information about NASDAQ and the investor should simply remain fully allocated; trying to act on small credit wobbles would only generate whipsaw. The picture changes once stress builds. A sharp widening of CDX is, first and foremost, a high-confidence confirmation that an equity drawdown is genuinely underway rather than market noise, since it coincides with NASDAQ declines in roughly 85 to 91 percent of cases, and, because a meaningful portion of the decline typically still lies ahead when it fires, it is a reasonable trigger to begin trimming exposure. How much to trim is informed by a second piece of information the credit market supplies: the magnitude of the CDX move scales with the eventual depth of the drawdown (a correlation of about −0.85), so a violent credit dislocation argues for a larger reduction in exposure than a mild one, even though the ultimate size of the move cannot be known at the first signal.

The most valuable signal arrives at the point investors find hardest to judge: the bottom. A decisive tightening of CDX in the high-stress regime has reliably preceded a NASDAQ recovery, and requiring a confirmed tightening, a second such move within about a month, sharpens it considerably, raising both the average 20-day gain and the hit rate well above those of a single move. Under episode-clustered inference the confirmed signal is statistically significant, and it survives the most demanding test we apply: holding the COVID crash and the 2022 bear market in the in-sample period and validating only on the ordinary corrections of 2023 to 2026, the confirmed signal still delivered an average 20-day gain of about 5 percent, positive in roughly five of every six episodes. This is where credit's informational edge is genuinely forward-looking, because risk appetite returns to the more liquid credit market before it returns to equities; it is the signal to rebuild the equity allocation that was trimmed during the decline. The episode that exemplifies the complete arc is the March 2020 dislocation, when the CDX spread peaked on the day of the NASDAQ trough and compressed sharply as policy support was announced, with equities recovering shortly thereafter.

Across eleven fully observed corrections spanning 2011 to 2026, this playbook would have engaged during every episode, scaled to severity through the credit move's magnitude, and flagged the recovery in the high-stress cases. The contribution of the study is therefore less a trading rule than a disciplined, evidence-based way to think about credit as an allocation overlay for equities: when to hold, when and how much to trim, and when to rebuild. Two limitations frame the result honestly. The strongest recovery evidence rests on a modest number of genuine high-stress episodes, so its precision should be expected to soften in any future sample, even though the out-

of-sample test already addresses this concern directly; and judging in real time whether an unfolding shock is regional or systemic, which governs how deep the drawdown becomes, remains difficult, even though CDX magnitude helps quantify it as the move develops. Promising extensions include applying the velocity-regime framework to CDX High Yield, which may lead earlier in credit-driven episodes, and to iTraxx Europe against European equity benchmarks, where the same credit-to-equity channel should operate. For an investor committed to holding NASDAQ over the long run, the message is practical and specific: let CDX confirm and size the defensive trim through a correction, and let a high-stress credit tightening signal the moment to put risk back on.

A natural and more ambitious extension, which we leave to separate work to avoid diluting the velocity-regime focus of this study, is a multivariate model that embeds CDX within a broader cross-asset stress system. Equity-implied volatility (VIX) and Treasury-implied volatility (the MOVE index) are the obvious additions, since all three spike together in stress and the Merton framework links equity volatility directly to credit spreads. Such a model would isolate the incremental information CDX carries about NASDAQ beyond what equity- and rate-volatility already reveal, the central robustness question for any credit-equity leadership claim. Existing multivariate work with CDX has taken different routes: Byun (2025) extracts a systemic credit risk premium from CDX tranches and shows it prices US equity risk, and related affine-model studies examine the joint dynamics of CDX and S&P 500 options, but none combines a velocity-defined regime structure with cross-asset volatility controls. A further promising ingredient is the option-implied Credit VIX family (VIXIG and VIXHY), introduced by Cboe in October 2023, which provides a forward-looking analogue to our realized-velocity measure and could serve as a leading complement once a sufficient history accumulates. Combining velocity-defined credit regimes with VIX, MOVE, and Credit VIX in a single multivariate framework is, to our knowledge, an open problem and a promising direction for future research.

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